

Question	Answer	Marks
5(a)	amplitude of carrier (wave) varies	B1
	variation in synchrony with displacement of information signal	B1
5(b)(i)	wavelength = $(3.0 \times 10^8)/(900 \times 10^3)$ = 3.3×10^2 m	A1
5(b)(ii)	amplitude varies (continuously) between a maximum and a minimum	B1
	variations repeat 5000 times each second or variations repeat every 0.2 ms or variations above and below 4.0 V	B1
5(b)(iii)	10000 Hz	A1
5(c)(i)	Any two from: • (orbit is) above the Equator • (orbit is) from west to east/same direction as Earth's rotation • orbit is circular/orbit has a particular radius	B2
5(c)(ii)	1. minimal reflection/absorption/attenuation by <u>atmosphere</u> or maximum penetration of/transmission through <u>atmosphere</u>	B1
	2. uplink signal is greatly attenuated/must be greatly amplified	B1
	prevents downlink signal swamping the uplink signal	B1

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$C(eV)$	work done per unit charge	D1